PROJECT 2

INTELLIGENT LEARNING ANALYTICS & AGENTIC AI STUDY COACH

From Predictive Analytics to Personalized AI Coaching

PROJECT OVERVIEW

This project involves the design and implementation of an **AI-powered learning analytics system** that evolves into an Agentic AI Study Coach.

In **Milestone 1**, the system analyzes student learning data using classical machine learning techniques to predict performance, identify learning gaps, and generate study recommendations.

In **Milestone 2**, the same system is extended into an agent-based AI application that autonomously reasons about student progress, retrieves learning resources, generates personalized study plans, and adapts recommendations over time.

The project emphasizes progressive use of machine learning to neural networks and then LLMs, agentic AI design, and deployment of a publicly accessible AI application.

CONSTRAINTS & REQUIREMENTS

TEAM SIZE	API BUDGET
3–4 Students	**Free Tier Only**
FRAMEWORK	HOSTING
LangGraph (Recommended)	**Mandatory**

APPROVED TECHNOLOGY STACK

LLMS (MILESTONE 2)

- Open-source models
- Free-tier APIs

AGENT FRAMEWORK

- LangGraph
- Chroma / FAISS (RAG)

ML & ANALYTICS (MILESTONE 1)

- Scikit-Learn (regression, clustering)
- Neural Networks (MLP - Optional)
- Pandas/NumPy (Preprocessing)

UI FRAMEWORK

- Streamlit
- Gradio

HOSTING PLATFORM (MANDATORY)

- Hugging Face Spaces
- Streamlit Community Cloud
- Render (Free Tier)

WARNING: Localhost-only demonstrations will not be accepted.

MILESTONE 1: ML-BASED LEARNING ANALYTICS SYSTEM

MID-SEM SUBMISSION

Objective: Develop a data-driven learning analytics system using classical ML + basic NLP that analyzes student performance data, identifies strengths and weaknesses, and generates rule-based or ML-based study recommendations.

Functional Requirements:

- Analyze student performance data (Quiz scores, Topic completion).
- Predict student performance and classify learners.
- Generate study recommendations.

TECHNICAL REQUIREMENTS (ML)

- **Preprocessing:** Missing values, Scaling.
- **Supervised:** Logistic/Linear Regression.
- **Unsupervised:** K-means clustering for grouping.
- **Evaluation:** Accuracy, Precision, RMSE.

INPUTS & OUTPUTS

- **Input:** Student CSV (Scores, Time spent).
- **Output:** Pass/Fail Prediction, Learner Category (At-risk/High-perf).
- **Recs:** Text-based study recommendations.

MID-SEM DELIVERABLES

- ML pipeline implementation.
- Model evaluation results.
- Feature engineering explanation.

- Hosted UI with upload & dashboard.

MILESTONE 2: AGENTIC AI STUDY COACH

END-SEM SUBMISSION

Objective: Transform the analytics system into an autonomous AI Study Coach that reasons for student performance, plans personalized learning paths, retrieves external learning content, and maintains session memory.

Functional Requirements:

- Goal understanding & Multi-step reasoning.
- Tool usage: Web search, Content summarization.
- Memory: Track progress across sessions.
- Interactive conversational interface.

TECHNICAL REQUIREMENTS (AGENTIC)

- **Framework:** LangGraph (Workflow & State).
- **RAG:** Retrieval with Chroma/FAISS.
- **Reasoning:** Chain-of-thought prompting.
- **Memory:** Session-based persistence.

STRUCTURED OUTPUT

- **Diagnosis:** Learning gaps analysis.
- **Plan:** Personalized strategy & Weekly goals.
- **Resources:** External URLs (Tutorials).

END-SEM DELIVERABLES

- Publicly hosted application link.
- Agentic workflow implementation.
- Interactive Study Coach UI.
- Github Repository.
- Demo Video (5-7 mins).
- **Extension (Pick 1):** Quiz Gen, Adaptive Difficulty, PDF Export, or Multi-student Dashboard.

Final Artifacts: Hosted Link, GitHub Repo, Demo Video.

EVALUATION CRITERIA

PHASE	WEIGHT	CRITERIA
Mid-Sem (Milestone 1)	25%	• Accuracy of ML Models (Prediction & Clustering) • Quality of feature engineering • Dashboard usability & Visualization quality
End-Sem (Milestone 2)	30%	• Quality of personalized study plans • Agent reasoning & state management • Relevance of retrieved resources (RAG) • User Experience & Deployment reliability